

Assignment 2. P+7 (Oulipian language modelling)

GPT-2 ranks every possible next word by probability, and the value of x determines how far down that ranking the replacement word comes from. Most of the probability is concentrated in a few top guesses, while the rest trails off across thousands of unlikely words. A smaller x therefore stays close to the model's first choices, producing words that are predictable, safe and often clichéd. A larger x reaches further down the list, producing results that are more random and often funnier.

My two versions illustrate this difference. In P+7, the replacements remain predictable: the new words stay within the poem's themes of nature and the seasons, such as "chill," "leaves," and "summer." At rank 7, GPT-2 is still close to its obvious guesses, and the poem's own context keeps pulling it back to the same vocabulary; "winter" even appears four times. P+34, by contrast, offers more random and unexpected choices. Some lines become witty: "cold a long spring" creates a small contradiction, and "the nothing that happens" gives Stevens' ending a new philosophical twist. However, the poem also begins to break down, with fragments such as "m," "ch" and "w," which shows that further down the ranking, the model's suggestions become less like real words.

To replace every noun in the poem rather than only the last word of each line, I would first identify the nouns using a Python library that tags each word's part of speech, such as spaCy. Then, for each noun, I would give GPT-2 the text that comes before it and take its seventh-ranked prediction. To keep the sentences grammatical, I would skip any candidates that are not nouns, counting only nouns until I reach the seventh. One limitation is that GPT-2 reads only from left to right, so it cannot consider the words that follow the noun it is replacing.

To summarize, x acts as a dial between predictability and chaos. Experimenting with it showed me that GPT-2 does not understand the poem; it simply calculates which words are likely, and the most interesting results appear when we deliberately choose the unlikely ones.